

REVISION: *Apis dorsata* Projected Distribution in Southeast Asia Under Moderate (SSP245) and High (SSP585) Emissions by 2050 and Its Socio-Economic Impacts

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I. INTRO

The giant honey bee (*Apis dorsata*) is a wild honey bee species, native to South and Southeast Asia. *Apis dorsata* plays an important ecological role as a pollinator. This species contributes to ecosystems by supporting plant reproduction and maintaining interactions between plants and other organisms. Through pollination networks, *Apis dorsata* helps maintain ecological stability and resilience of ecosystems.

Apis dorsata was chosen as the focal species because it is native to Southeast Asia, and gives a broader view of climate change study since this species is typically found in the wild. Unlike honey bee species like *Apis mellifera* that are usually maintained through human intervention, *A. dorsata* depends entirely on natural environments for nesting, foraging, and reproduction. Therefore, changes in its distribution pattern can provide important insights into how climate change may affect wild pollinator populations.

Species Distribution Modeling (SDM) is a useful approach to predict potential habitat suitability. This is done by relating species occurrence records with environmental variables. This model is meant to predict the distribution of *Apis dorsata* by the year 2050, by mapping suitability habitats under two different emission scenarios, SSP245 (moderate emissions) and SSP585 (high emissions). Shared Socioeconomic Pathway (SSP) is a set of future climate scenarios, commonly used in climate modeling to project future climate conditions. SSP245 is considered moderate, with the society continues developing but emissions do not decline quickly enough to avoid warming. SSP585 is considered as high emissions, with society relying heavily on fossil fuels and greater greenhouse gas emissions.

The use of an ensemble of Species Distribution Modeling is widely regarded as best practice for projecting species responses to climate change because different statistical algorithms capture distinct facets of the species-environment relationship, exhibit different sensitivities to data quality, and can yield divergent projections when applied independently (Hao *et al.*, 2019; Ruzzier *et al.*, 2024). Applied to *Apis dorsata*, this principle becomes particularly relevant because the species occupies habitat ranging from deciduous dipterocarp to evergreen rainforest, and undergoes seasonal migration along major riparian corridors. A single algorithm is therefore unlikely to adequately represent the simultaneous influence of temperature seasonality, river proximity, and forest cover on colony presence.

Comparable recent precedents confirm the approach: a biomod2 ensemble has been used for *Apis cerana* across southern China (Chen *et al.*, 2026), species distribution modeling has been used to project *A. dorsata* range shifts under SSP245 and SSP585 in Pakistan (Khan *et al.*, 2025). Accordingly, this independent study aims

to identify potential areas of habitat loss and gain under future climate conditions using biomod2 ensemble, and provide insights into how climate change may influence the distribution of a key native pollinator species in Southeast Asia, as well as the socio-economic impacts caused by it.

II. METHODS

2.1 SPECIES OCCURRENCE DATA

Occurrence records for *Apis dorsata* were downloaded from the Global Biodiversity Information Facility (GBIF) using the rgbif package (Chamberlain & Boettiger, 2017), restricted to records with coordinates and no flagged geospatial issues. This returned 9,169 records.

Records were cleaned to remove duplicate coordinates and common sources of geographic error using CoordinateCleaner (Zizka *et al.*, 2019), which flags records falling on country/province centroids, biodiversity institution addresses, and other known artefact locations. After removing 1,806 duplicate coordinate pairs and a further 556 records flagged by these tests, 6,807 records remained.

Because GBIF records are unevenly distributed in space, often clustered around cities, roads, and frequently visited sites, occurrence data were spatially thinned to reduce this sampling bias before model training, using spThin (Aiello-Lammens *et al.*, 2015). A minimum nearest-neighbour distance of 10 km was applied, chosen to roughly match the spatial resolution of the climate data used later while still preserving meaningful spatial variation in occurrence records across the study region. This step retained 2,124 records. Only a single thinning replicate was run; spThin's authors recommend multiple replicates (e.g. 100) to account for the algorithm's stochastic point-selection, so a degree of variability attributable to thinning alone was not formally assessed and is noted as a limitation.

Out of all 2,124 spatially thinned records, 988 fell outside of the Southeast Asian study area and 102 further records were dropped due to missing environmental data at their location, leaving a final training set of 1,034 records.

2.2 STUDY AREA AND ENVIRONMENTAL PREDICTORS

The study area was defined as continental and insular Southeast Asia (90–150°E, 15°S–30°N). Nineteen bioclimatic variables were obtained from WorldClim v2 at 10 arc-minute resolution (approximately 18.5 km at the equator; Fick & Hijmans, 2017). This resolution was chosen to balance regional-scale climatic signal against the spatial precision of the occurrence data (much of which carries coordinate uncertainty on the order of hundreds of metres to kilometres) and against the computational demands of training an ensemble of models and projecting them under multiple future scenarios across the full study extent.

In addition to climate, three further groups of predictors were compiled, reflecting known drivers of *A. dorsata* nesting and foraging habitat:

- 1) Topography/elevation(WorldClim v2), with slope and aspect derived from it, capturing terrain steepness and orientation that influence nest-site exposure and microclimate.

- 2) Forest cover/percentage tree cover and forest loss year, from the Hansen Global Forest Change dataset (Hansen *et al.*, 2013), reflecting the species' reliance on tall trees for nesting.
- 3) Hydrology/distance to the nearest river, derived from HydroSHEDS river network data (Lehner *et al.*, 2008).

All layers were cropped to the study extent, masked to land area only (to avoid non-terrestrial cells influencing subsequent statistical checks), and resampled onto the climate grid. Continuous variables (elevation, tree cover) were resampled using bilinear interpolation; the categorical forest-loss-year layer was resampled using nearest-neighbour assignment, which preserves discrete category values rather than averaging them.

2.3 MULTICOLLINEARITY ASSESSMENT AND PREDICTOR SELECTION

Because several algorithms used in the modeling ensemble such as GLM and GAM are sensitive to collinearity among predictors, redundant variables were screened out in two stages, following common practice in species distribution modeling (Dormann *et al.*, 2013):

1. Pairwise Pearson correlations were calculated from a random sample of 10,000 raster cells, and any variable pair exceeding $|r| = 0.7$ was flagged.
2. Variance Inflation Factor (VIF) analysis was then applied to the retained set, with any variable having $VIF > 10$ removed. This step captures multi-variable redundancy that pairwise correlation alone can miss, since a variable can be well predicted by a *combination* of several others without being strongly correlated with any single one. The threshold of $VIF = 10$ follows conventional regression diagnostic guidance (Kutner *et al.*, 2004).

This process was applied first to the 19 bioclim variables alone, retaining seven (BIO2, BIO4, BIO5, BIO13, BIO14, BIO18, BIO19), and then re-applied to the complete 13-variable stack once topography, forest, and hydrology layers were added, since collinearity between climate and non-climate predictors could not be assessed until all layers were combined. This second pass excluded elevation ($VIF > 10$), leaving a final set of 12 predictors: BIO2, BIO4, BIO5, BIO13, BIO14, BIO18, BIO19, Slope, Aspect, Tree Cover, Forest Loss, and Distance to River.

2.4 PRESENCE DATA PREPARATION

Environmental values were extracted at each of the 2,124 thinned occurrence coordinates. Records falling outside the predictor raster's extent (988 records, largely *A. dorsata* occurrences from South Asia and other parts of the species' broader native range that fall outside the Southeast Asian study region defined here) were excluded, leaving 1,136 records. A further 102 records were excluded due to missing environmental values at their location, arising from two distinct causes: (1) coastal or small-island points falling on raster cells masked out during land/ocean separation, and (2) edge-effect gaps in the slope and aspect layers, an artefact of how terrain derivatives are calculated at the margins of the elevation raster. The latter were repaired using focal-mean interpolation (a 3×3 moving window average applied only to NA cells) prior to the final extraction, since these represent a computable value rather than genuinely missing information; the former (masked ocean/coastal cells) were not repaired, since no meaningful terrestrial environmental value exists there. This yielded a final training set of 1,034 occurrence records.

2.5 PSEUDO-ABSENCE GENERATION AND DATA FORMATTING

Because occurrence-only records provide no explicit information on unsuitable locations, pseudo-absence (background) points were generated to allow use of the presence/pseudo-absence classification algorithms below, following the general approach and rationale reviewed in Barbet-Massin *et al.* (2012). Data were formatted

using biomod2 (Thuiller *et al.*, 2009), generating three independent pseudo-absence sets of 5,000 randomly sampled background points each. Multiple independent pseudo-absence sets were used, rather than a single set, to reduce the influence of any one arbitrary background sample on the fitted models.

2.6 MODEL TRAINING AND EVALUATION

Species distribution models were trained using biomod2, which implements a common ensemble-of-algorithms framework for SDMs (Thuiller *et al.*, 2009). Six algorithms were used: generalized linear models (GLM), generalized additive models (GAM), generalized boosted models (GBM), random forest (RF), extreme gradient boosting (XGBoost), and maximum entropy modeling via the MAXNET implementation. For each of the three pseudo-absence sets, three repeated random 80/20 train–test splits were used for cross-validation, giving a total of 3 (pseudo-absence sets) $\times$ 3 (cross-validation repeats) $\times$ 6 (algorithms) = 54 individual models. Model discrimination performance was evaluated using the True Skill Statistic (TSS) and the area under the receiver operating characteristic curve (AUC; Allouche *et al.*, 2006; Fielding & Bell, 1997), and permutation-based variable importance (5 permutations per model) was recorded for each model at training time.

Calibration and validation performance were compared across algorithms to check for overfitting, a substantial drop in an algorithm's relative ranking between calibration and validation performance indicates it is fitting noise in the training data rather than a generalizable signal. Algorithms and pseudo-absence/cross-validation repeats meeting a validation TSS threshold of 0.40, a threshold commonly used in biomod2-based studies to distinguish models with acceptable discriminatory power from poor performers, were retained for the ensemble model described below; the remainder were excluded from ensemble construction (though their outputs remain available for reference).

2.7 VARIABLE IMPORTANCE AND RESPONSE CURVES

Permutation-based variable importance scores across all retained models were summarised by mean and standard deviation per predictor. Because random forest is known to be prone to inflating apparent importance for certain variable types and had already shown a strong calibration–validation performance gap, variable importance and response curve interpretation were additionally computed excluding RF, to check whether its inclusion was distorting the overall ranking. Response curves were generated for the top-ranked predictors to characterise the direction and shape (linear, unimodal, or threshold-like) of each variable's relationship with predicted suitability.

2.8 ENSEMBLE MODEL BUILDING

An ensemble model was built from the pool of individual models meeting the TSS > 0.40 validation threshold, using both an unweighted mean (EMmean) and a performance-weighted mean (EMwmean, weighted by each contributing model's TSS score) consensus method. Consensus/ensemble approaches of this kind are widely used in SDM studies because they tend to average out idiosyncratic errors of

individual algorithms and produce more stable predictions than any single model (Marmion *et al.*, 2009; Thuiller *et al.*, 2009; Araújo *et al.*, 2007). A single TSS-optimised threshold was used to convert continuous habitat suitability predictions into a binary suitable/unsuitable classification, applied consistently across current and future projections.

2.9 HABITAT SUITABILITY PROJECTIONS WITH FUTURE CLIMATE SCENARIOS

The ensemble model was projected onto the current climate/environment layers to generate a present-day habitat suitability surface, then thresholded to a binary suitable/unsuitable map. To assess potential future range shifts, the same set of trained models was projected onto future climate conditions under two Shared Socioeconomic Pathway scenarios for the 2041–2060 period (referred to here as "2050"): SSP245 (a moderate-emissions pathway) and SSP585 (a high-emissions pathway), following the SSP/RCP scenario framework adopted for CMIP6 (O'Neill *et al.*, 2016). Future bioclimatic layers were cropped and resampled onto the same grid as the current predictors. Topographic, forest, and hydrological predictors were held constant at current values in the future projections, since dynamic future projections for these layers were not available; this is a simplifying assumption and is noted as a limitation, since forest cover and river networks are themselves likely to change over the same time horizon under land-use and climate pressures.

Change between current and future binary suitability maps was classified per grid cell as habitat loss, stable habitat, or habitat gain, and summarised both as the percentage of grid cells in each category and as total area (km²) under each scenario, using each cell's true geodesic area rather than a flat per-cell count to account for the fact that grid cell area varies with latitude in an unprojected coordinate system.

III. RESULTS AND DISCUSSIONS

3.1 MODEL PERFORMANCE

Of the 54 individual models trained (6 algorithms $\times$ 3 pseudo-absence sets $\times$ 3 cross-validation repeats), performance varied substantially by algorithm (Table 1). XGBoost, GAM, and MAXNET showed the best and most consistent validation performance (mean validation TSS 0.46, 0.44, and 0.43 respectively). Random forest showed a striking discrepancy between calibration (mean TSS = 1.00, i.e. perfect fit to training data in every run) and validation (mean TSS = 0.37, the lowest of all six algorithms). That shows a clear overfitting signature, since a model that fits its training data perfectly but performs worst on held-out data is learning noise specific to that training set rather than a generalizable relationship. GLM was the weakest algorithm on validation performance (mean TSS = 0.38).

Table 1. Mean calibration and validation performance by algorithm across all pseudo-absence sets and CV repeats

Algorithm	TSS (calibration)	TSS (validation)	AUC (calibration)	AUC (validation)
XGBoost	0.59	0.46	0.86	0.80
GAM	0.51	0.44	0.83	0.79
MAXNET	0.50	0.43	0.83	0.80
GBM	0.48	0.41	0.80	0.77
RF	1.00	0.37	1.00	0.85
GLM	0.42	0.38	0.77	0.75

Of the 54 individual models, 30 met the $TSS > 0.40$ validation threshold used to build the ensemble. These were unevenly distributed across algorithms: XGBoost contributed 9 of its 9 runs, GAM 8 of 9, MAXNET 7 of 9, GBM 4 of 9, and both GLM and RF contributed only 1 of 9 runs each, consistent with GLM's consistently weak performance and RF's overfitting.

With RF excluded from the importance calculation, BIO4 (Temperature Seasonality) was by far the strongest predictor of habitat suitability, with a mean permutation importance roughly four times higher than the next-ranked variable (Figure 1). Distance to river was the second strongest predictor, followed by tree cover, and then four precipitation-related bioclim variables (BIO18, BIO13, BIO19) and maximum temperature of the warmest month (BIO5) at comparatively low and similar importance. Slope, minimum temperature seasonality-related variables (BIO14, BIO2), aspect, and forest loss contributed the least.

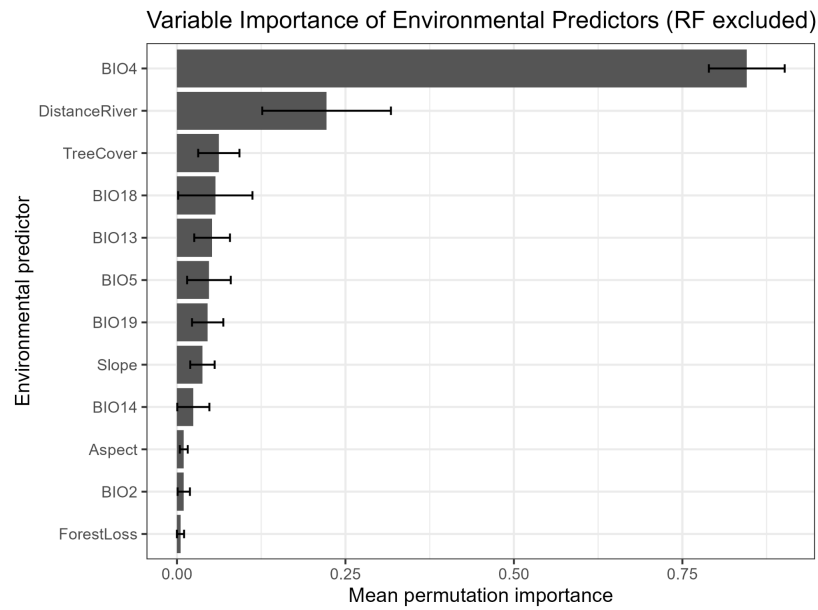


Figure 1. Variable importance of environmental predictors for *Apis dorsata* habitat suitability; permutation importance (mean $\pm$ SD) of 11 predictors averaged across multi-algorithm ensembles (excluding RF). BIO4 (Temperature Seasonality) was the dominant variable ($\sim$ 0.82), followed by Distance to River ($\sim$ 0.22), all other predictors contributed <0.10 indicating that the species' distribution is largely governed by a single climate dimension

Response curves (Figure 2) show BIO4 (Temperature Seasonality) experiences a strongly decreasing relationship, suitability is highest at low seasonality and drops toward zero above roughly 500–600 (standard deviation $\times$ 100 of monthly temperature), indicating the species is largely restricted to climatically stable, non-seasonal tropical environments. The overwhelming dominance of BIO4 (temperature seasonality) is consistent with *A. dorsata* being a lowland tropical specialist with limited tolerance for temperature fluctuation, a pattern that matches the species' known biology, since colonies build large, exposed, single-comb nests that are more vulnerable to microclimate extremes than the enclosed cavity nests of *A. mellifera* or *A. cerana*.

Apis dorsata spans latitudes from $\sim$ 28°N down to $\sim$ 10°S and crosses the equator, so it is exposed to very different annual temperature regimes from strongly seasonal continental climates in northern India and Myanmar to equatorial climates in Borneo and Sumatra. BIO4 coming out as the top predictor through permutation importance is consistent with the species' known migratory strategy, in which colonies move between lowland and highland or seasonal and aseasonal flowering zones (Robinson, 2012; Robinson, 2021). A high degree of permutation importance of temperature seasonality has been mirrored in other tropical and subtropical studies, where BIO4 and BIO15 typically replace BIO1 as the leading variable for migratory and wide-ranging taxa (Ganglo *et al.*, 2017).

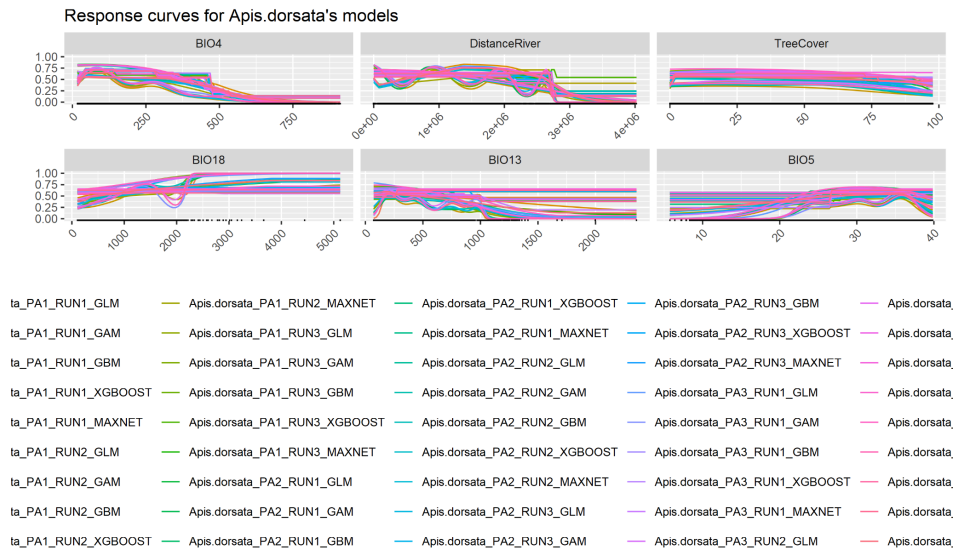


Figure 2. Response curves of *Apis dorsata* to the leading environmental predictors; response curves show probability of presence across the gradient of each major predictor. These curves define the climatic niche of *A. dorsata*

Distance to River shows a broadly flat to declining relationship, with suitability relatively stable at shorter distances and falling off beyond roughly 2.5–3 million metres, consistent with a preference for areas within reach of water sources rather than a narrow optimum distance. *A. dorsata* is known to forage up to several hundred meters from its nest, but its documented migratory stopover sites consistently occur along major rivers (Robinson, 2012). Robinson's long-term work at the Pai River showed that migrating colonies aggregate annually in a mango orchard beside the river, where abundant water and food support life-cycle stages (Robinson, 2021). Thus, Distance to River represents access to riparian corridors used during migration, water collection at the nest, and floral resources along forests.

Tree Cover shows a gradual decline from a plateau at low-to-moderate cover toward lower suitability at very high tree cover values, though the effect is comparatively weak. The prominence of distance to river and (to a lesser extent) tree cover as the next most important predictors is also ecologically coherent rather than a modeling artefact: *A. dorsata* is known to nest almost exclusively in large trees, and colonies are strongly associated with proximity to forest and water resources needed for colony thermoregulation and foraging. A study of coffee-pollinating bees in Kodagu, India, found that *A. dorsata* colonies were "largely restricted to forests due to a requirement for large trees" for nesting (Krishnan *et al.*, 2012), directly consistent with tree cover and (indirectly, via riparian forest) distance-to-river both emerging as meaningful predictors here, independent of climate.

BIO18 (Precipitation of Warmest Quarter) shows an increasing relationship up to roughly 2,000–2,500 mm before plateauing, suggesting wetter warm-season conditions favour suitability up to a point. This confirms that *A. dorsata* depends on massive, predictable monsoon rainfall during the warmest quarter (Itioka *et al.*, 2001; Raffiudin *et al.*, 2024).

BIO13 (Precipitation of Wettest Month) shows flatter, more variable relationships with less consistent direction across the contributing models, indicating a weaker and less

well-resolved influence on predicted suitability than the top two predictors. BIO13 peaks around 1,000-2,000 mm, while BIO5 suitability rises with temperature up to a peak around 30°C before declining. The BIO5 optimum near 30°C matches temperatures at which colonies maintain foraging activity without overheating stress, while the sharp decline at higher values likely reflects on thermal limits to drone production and brood development.

Response curve shape matters for adaptation. The shallow gradient of Tree Cover curve implies that moderate canopy restoration could retain much of the suitability value even where primary forest has been lost. Conversely, the steep BIO4 and BIO18 curves signal low adaptive redundancy. This means that once seasonality or monsoon precipitation crosses a threshold, suitability drops significantly. For vulnerable countries where these thresholds are being approached, proactive conservation of existing forest fragments and riparian connectivity is far more cost-effective than future restoration (Millard *et al.*, 2023; Pang *et al.*, 2024).

3.2 CURRENT HABITAT SUITABILITY

The current suitability map (Figure 3) shows a broadly contiguous suitable zone running through mainland Southeast Asia, corridor spanning Myanmar, Thailand, Laos, Cambodia, and Vietnam, along with the Malay Peninsula, Sumatra, much of the Philippine archipelago, and coastal fringes of Borneo. Notably, the interior of Borneo (Kalimantan) and much of Sulawesi appear largely unsuitable, despite these areas being part of the species' broader known range.

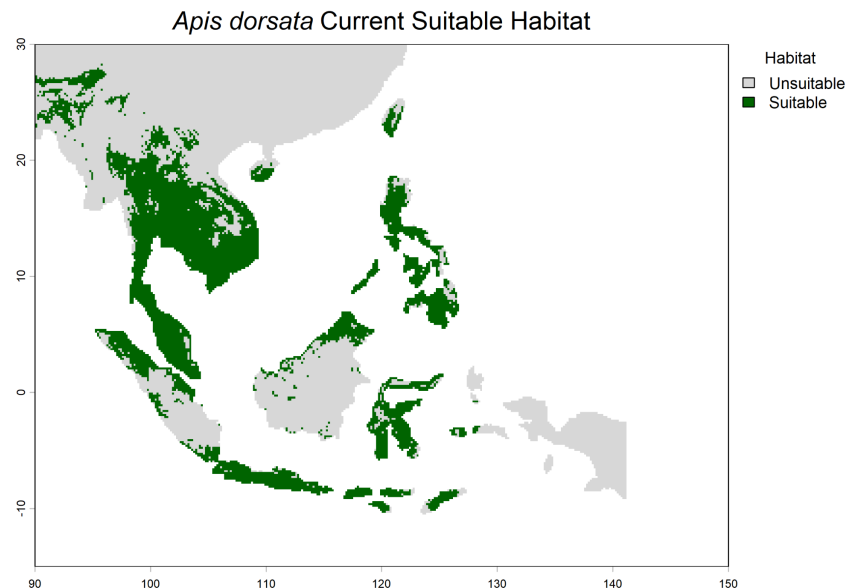


Figure 3. Current climatically suitable habitat for *Apis dorsata* across Southeast Asia

The current suitable habitat overlaps with some of the most pollinator-dependent smallholder economies: rice-longan system in Thailand (Yotapakdee *et al.*, 2024), oil palm and durian in Sumatra and Borneo (Raffiudin *et al.*, 2024; Sheherazade *et al.*, 2019). The wild-pollinator value of nearby forest alone has been estimated at ~RM 6.5 million (~USD 1.4 million) annually for an individual sub-district in Malaysia (Adnan *et al.*, 2021),

illustrating how the same pixels that are climatically suitable are also economically productive. Therefore, any future contraction should be read not only as a biodiversity loss, but also as a loss of ecosystem-service value.

3.3 FUTURE CLIMATE SCENARIOS

Aggregated across the whole study area, projected habitat change by 2050 was modest in relative terms under both scenarios (Figure 4): under SSP245, 94.2% of the study area is projected to remain stable, with 3.0% loss and 2.8% gain; under SSP585, 92.9% remains stable, with a larger 4.0% loss and 3.0% gain. The higher-emissions scenario therefore shows both greater habitat loss and a slightly larger stable-habitat erosion than the moderate-emissions scenario, as expected.

This aggregate stability, however, masks a clear spatial pattern (Figure 5): habitat loss is concentrated in the northern and upland margins of the mainland corridor particularly the Myanmar–Thailand–Laos border region and northeastern India, while habitat gain is concentrated in the equatorial insular zone, especially southern Sumatra and, more prominently under SSP585, the interior of Borneo. This pattern is consistent with a climate envelope shifting away from currently marginal, more temperature-seasonal peripheral zones and into currently under-suitable equatorial zones as warming reduces seasonality differences there.

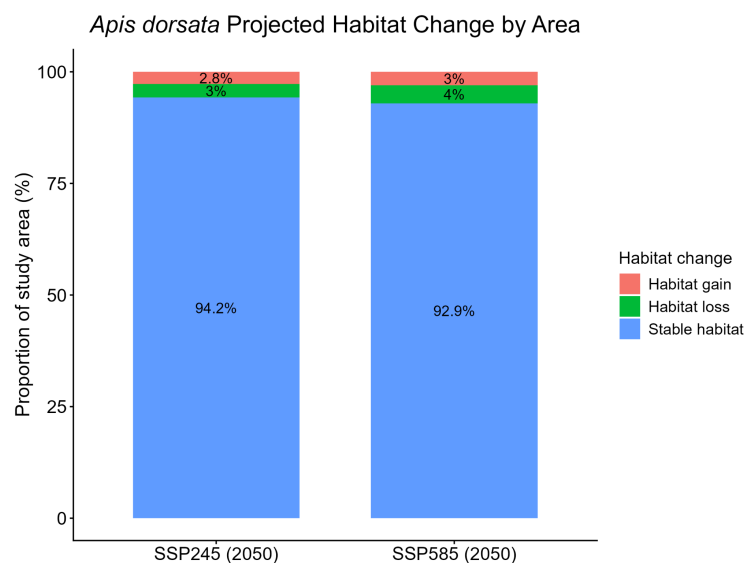


Figure 4. Projected habitat change by area (km²) by 2050 in Southeast Asia

Compared to the percentages shown in Figure 4, the combined loss in all Southeast Asia study areas is more concerning (Table 2). SSP585 produces a net loss more than four times larger than SSP245 (−64,504 km² vs. −14,776 km²), because both loss and gain grow under the higher-emissions pathway, but loss grows faster. Percentage rounding in a single stacked bar chart can understate how much worse the higher-emissions scenario really is; the underlying area figures should be reported alongside the percentages.

Table 2. Study area-wide total net change under SSP245 and SSP585 by 2050

	SSP245 (2050)	SSP585 (2050)
Total habitat loss	191,558 km ²	257,564 km ²
Total habitat gain	176,782 km ²	193,060 km ²
Net change	-14,776 km²	-64,504 km²

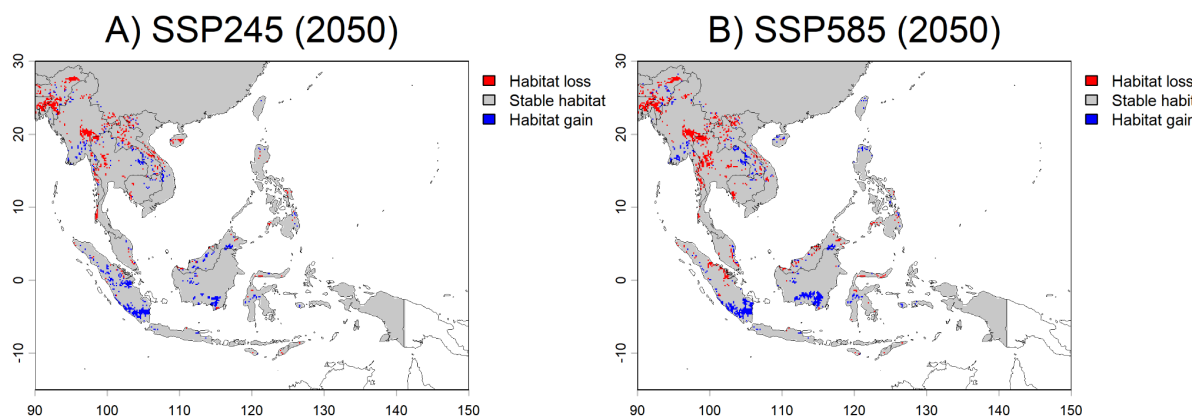


Figure 5. Geographic distribution of habitat gains, losses, and stable areas for *Apis dorsata* under SSP245 (A) and SSP585 (B) by 2050. Red pixels indicate projected habitat loss, blue pixels indicate projected habitat gain, and uncolored areas indicate stable suitability

The “shrinkage-and-shift” pattern shown in Figure 5 has been documented for *A. dorsata* in Pakistan under both SSP245 and SSP585 (Khan *et al.*, 2025) and for *A. cerana* across southern China (Chen *et al.*, 2026). As both maps shown, habitat loss (in red pixels) dominate northern South Asia and the central dry zone of Myanmar, while habitat gain (in blue pixels) cluster along the southern margins of current suitable area, particularly Sumatra, Java, and southern Borneo. Under SSP245, stable habitat still covers most of Sundaland and Indochina. Under SSP585, the loss area expands into mainland Southeast Asia, and the gain zones are smaller and more fragmented, indicating that new areas only marginally compensate for broader losses.

Since *A. dorsata* is a migratory species, its capacity to track climate is greater than that of many non-migratory taxa. However, the persistence of those migratory routes depends on stopover sites, which are typically riverine and heavily modified by agriculture and infrastructure (Robinson, 2012; Robinson, 2021). Therefore, even regions showing net habitat gain will only deliver their pollination potential if riparian connectivity is preserved and issued not only as a climate issue but also policy issue.

Reading approximate values from the top-10-countries chart under SSP585, Thailand (~50,000 km²), Myanmar (~48,000 km²), and India (~41,500 km²) show the greatest absolute projected habitat loss, followed by Indonesia (~30,000 km²), Malaysia (~19,000 km²), Laos and Vietnam (~16,000 km² each), and Bangladesh (~14,000 km²). Most countries show greater loss under SSP585 than SSP245, consistent with the more severe warming trajectory, with one notable exception: Laos shows *more* projected loss under the moderate SSP245 scenario (~20,000 km²) than under the higher-emissions SSP585 scenario (~16,000 km²). It could reflect a genuine non-linear response (e.g. increased precipitation under the higher-emissions pathway partially offsetting temperature-driven suitability loss in a country where the balance of limiting factors differs from its neighbours), or it could warrant a data-check before being reported as a genuine finding.

Thailand and Indonesia show the steepest escalation from SSP245 to SSP585. SSP585 adds a further ~10,000 km² of habitat loss in Thailand and ~20,000 km² in Indonesia relative to SSP245, while Philippines and Cambodia remain low in absolute terms but suffer complete or near-complete loss of their marginally suitable areas.

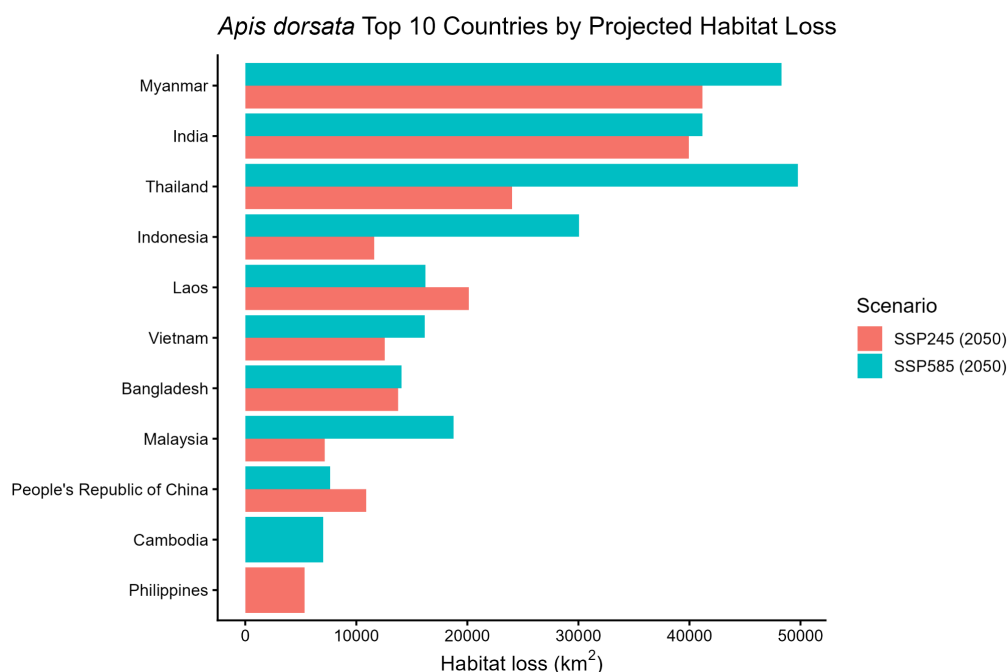


Figure 6. Projected habitat loss (km²) of suitable habitat for *Apis dorsata* by 2050 across top 10 affected countries under SSP245 and SSP585

This pattern aligns with the broader prediction that tropical lowlands, not temperate uplands, will see the steepest biodiversity losses under high-emissions pathways (Millard *et al.*, 2023). Myanmar and India's combined exposure is partly attributable to their large geographic extent within the species' core thermal envelope; when BIO4 shifts outside that envelope, the loss is correspondingly large in absolute terms. The disproportionate surge for Thailand under SSP585 mirrors recent ensemble-SDM findings for the sister species *Apis*

cerana in China, where a moderate scenario produced limited loss while SSP585 drove over 40% contraction (Chen *et al.*, 2026).

The headline number from the future scenarios, only 3–4% net habitat loss across the whole study area, could be read as reassuring in isolation. That would be a misleading takeaway. The spatial pattern shows the loss is not spread evenly; it is concentrated at the northern, upland margin of the mainland range (Myanmar–Thailand–Laos border zone, northeastern India), the area furthest from the equatorial, low-seasonality conditions the models identify as optimal. As an important side note, India and China's figures reflect only the fragments of those countries within the 90–150°E study extent (8% and 21% of their national territory respectively), not national totals, a point worth restating prominently anywhere these numbers are quoted outside this report, since the raw km² and even the normalized percentages above could otherwise be mistaken for country-wide statistics.

This is the expected signature of a climate-driven range contraction at a species' thermal margin, a pattern well documented in the broader climate change biology literature (Parmesan *et al.*, 2003; Chen *et al.*, 2011). Aggregating this into a single study-area-wide percentage obscures the fact that specific countries and specific populations, not the region as a whole evenly, bear the brunt of projected loss. A 3–4% regional loss composed mostly of decline concentrated in Thailand, Myanmar, and northern India (collectively well over 100,000 km² of projected loss under SSP585, per the country-level figures above) is a materially different conservation story than a uniform 3–4% loss spread evenly across all eleven countries in the study region.

3.4 SOCIO-ECONOMIC IMPLICATIONS: POLLINATION-DEPENDENT CROPS IN THE PROJECTED LOSS ZONE

Southeast and South Asia are regions where pollination-linked agricultural output represents a disproportionately large share of total crop value. The IPBES global pollinator assessment identifies South Asia and East Asia among the world regions with the greatest pollination-linked contribution to crop output, in the range of 5–15% of total crop value depending on region (IPBES, 2016), and a widely cited global estimate puts the total economic value of insect pollination at roughly € 153 billion annually (USD 175 billion), or 9.5% of the value of global food production (Gallai *et al.*, 2009). *A. dorsata* specifically has been documented as an economically significant pollinator of several crops grown across its range, including coffee, mango, coconut, and durian.

In southern Thailand, *A. dorsata* was recorded as the most frequent floral visitor to durian and, while fruit bats are generally the primary pollinator, *A. dorsata* was found to be an effective supplementary pollinator, and the dominant one in areas where bat pollinators are scarce, for the widely cultivated 'Monthong' cultivar (Bumrungsri *et al.*, 2009; Wayo *et al.*, 2018). Thailand is one of the world's largest durian producers and exporters, and shows some of the largest projected absolute habitat losses in this study's country-level results. *A. dorsata* also has been documented forming large seasonal migratory congregations directly within mango orchards (Robinson, 2012), indicating a close ecological association between the species and mango-growing landscapes in a country where mango is a significant fruit export crop. Additionally, bee pollination on longan farms yields up to 68 kg/rai, with a derived

economic value of ~13,080 THB/rai (USD 389/1600 m²) (Yotapakdee *et al.*, 2024). A loss of ~51,000 km² under SSP585 risks undermining these yields.

In Myanmar, rural honey hunting from wild *A. dorsata* colonies form a significant non-timber forest production stream, analogous to documented Sumbawa communities where honey accounts for 68% of household cash income and is described by 83% of respondents as essential (Schouten *et al.*, 2019). Habitat loss with the magnitude of ~46,500 km² under SSP585 therefore threatens rural livelihoods and the supply of forest-derived honey traded in domestic markets (Harianja *et al.*, 2023).

In Indonesia, 80% of its national honey supply originates from wild *A. dorsata* colonies in Sumbawa alone (Schouten *et al.*, 2019). Comparable village economies in Southwest Maluku have been documented to produce 1,500 L of forest honey annually, generating net household incomes exceeding IDR 169 million/year (USD 9,419/year) (Sahureka *et al.*, 2024). A loss of ~32,000 km² under SSP585 has a cultural and economic impact on honey-hunter communities.

In Kodagu, one of India's principal coffee-growing regions, *A. dorsata* accounted for 47–58% of floral visitors to coffee (*Coffea canephora*) and increased fruit set by roughly 50% relative to wind-pollination alone (Krishnan *et al.*, 2012). This is a direct, quantified demonstration of the species' contribution to a major regional cash crop. It's worth noting explicitly that Kodagu itself (~76°E) sits well outside this study's modeled extent (90–150°E), so this finding illustrates the species' broader economic role in India rather than mapping directly onto the northeastern India fragment captured in this study's projections, the underlying ecological dependency (forest-nesting *A. dorsata* pollinating agroforestry crops) is nonetheless directly relevant to how habitat loss elsewhere in the species' Indian range should be interpreted. More broadly, *A. dorsata* is reported as a contributing pollinator of coconut, black pepper, star fruit, macadamia, and cotton across South and Southeast Asia (Sihag, 2025). All crops grown at commercial scale within the countries showing the greatest projected habitat loss in this study (Thailand, Myanmar, India, Indonesia, Vietnam).

Taken together, this positions the projected habitat contraction not as an abstract biodiversity metric but as a plausible pressure on pollination services for specific, economically significant crops in specific countries, with coffee in India and durian/mango in Thailand as the clearest, best-evidenced examples given current literature. A natural next step, beyond the scope of the current SDM, would be to spatially overlay the projected habitat loss layers against maps of coffee, durian, and mango cultivation area within Thailand and India specifically, to quantify the overlap between projected *A. dorsata* habitat loss and areas of documented crop dependency, this would convert the current qualitative case into a spatially explicit vulnerability estimate.

V. CONCLUSION

By 2050, projected losses of climatically suitable habitat for *Apis dorsata* are concentrated in the northern and central portions of Myanmar, India, and Thailand, while modest gains appear in southern Sundaland. The dominance of BIO4 (Temperature Seasonality) in the model indicates that the species' distribution is constrained less by mean climate than by the predictable alternation of wet and dry seasons that triggers mass-flowering events on which it depends. Disturbance of this seasonality by anthropogenic climate change therefore poses a cascading risk to:

- 1) The pollination of forest-dependent and agricultural plant species,
- 2) The wild-honey economy on which rural communities in Indonesia, Myanmar, and parts of India depend, and
- 3) The cultural practices of traditional honey-hunters whose livelihoods are tightly coupled to colony migration along major river systems.

Since SSP585 dramatically elevates loss relative to SSP245, especially for Indonesia and Thailand, the results in this study align with global syntheses identifying the tropics, Southeast Asia in particular, as the geographical epicentre of pollinator-related agricultural risk under high-emission pathways. Therefore, conservation strategies should prioritize:

- 1) Protection of riparian stopover sites,
- 2) Tree-cover retention and restoration in the 0-40% canopy zone where the species' response curves remains relatively tolerant, and
- 3) Climate mitigation, given that the same forests sequester carbon, and support biodiversity as well as pollinator-dependent livelihoods.

VI. LIMITATIONS

Several limitations flagged during model development remain relevant to interpreting these results: occurrence thinning was performed with a single replicate rather than the recommended multiple repetitions; topographic, forest, and hydrological predictors were held static in the future projections despite these being plausibly dynamic over the same 2050 time horizon; the rectangular study extent excludes part of the species' broader native range (including much of India) and creates border-fragment artefacts for countries like India in the country-level breakdown; and the Laos anomaly noted above warrants a data-check before being treated as a confirmed finding.

VI. REFERENCES

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